

Homework — 1.5.2 Moral and Ethical Issues — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] Completed match-up (letters G — open source — and H — copyright infringement — are not used):

Term	Match
Automated decision making	A
Algorithmic bias	B
"Black box" system	C
Offshoring	D
Monopoly	E
Digital divide	F

Mark: 2 correct rows per mark (6 rows → 3 marks).

2. [2] Correct ordering:

Step	Order
The deployed system misidentifies people from the under-represented groups far more often.	5
Engineers train the model on this dataset.	2
A dataset of faces is collected that under-represents some ethnic groups.	1
The police force puts the system into use for identification.	4
The model learns facial features mainly from the over-represented groups.	3

Full sequence: *biased data collected* → *model trained on it* → *model learns the skewed patterns* → *system deployed* → *biased outcomes appear*. Mark: all five positions correct = 2; three or four correct = 1.

3. (a) [1] Any one: faster than manual reading / can process huge numbers of scans without fatigue / consistent results / acts as a safety net catching things a tired clinician might miss.

(b) [1] Any one: it is unclear who is **accountable/liable** if the system misses or wrongly flags a tumour; the system could be **biased** if trained on unrepresentative scan data; over-reliance could **de-skill** staff so errors go unquestioned; a wrong automated decision has serious consequences for the patient.

4. [2] Completed paragraph, in order of the blanks:

1. **monopoly**
2. **lock-in**
3. **source code**

4. vendor

Unused distractors: copyright, patent.

Mark: 2 correct blanks per mark (4 blanks → 2 marks).

5. [5] Levels-of-response "**discuss**" question — mark the **whole answer**, not point-by-point. Expect a balanced discussion across stakeholders with a justified conclusion.

Indicative content — - **FOR**: cheaper to run / saves council money; faster, 24/7 access for the digitally-confident; less paperwork; easier to update. - **AGAINST**: the **digital divide** excludes those without internet access, devices or skills — disproportionately the elderly, poor, rural or disabled, who may be unable to claim essential benefits; loss of face-to-face help; widening inequality. - **Stakeholders (aim 3+)**: the council/taxpayers · digitally-confident residents · digitally-excluded residents (elderly/poor/disabled) · staff whose roles change. - **Justified conclusion**: must take a clear, ideally *conditional* position, e.g. *"moving services online is justified to cut costs only if assisted/phone/in-person routes remain for those who cannot access them, so no one is excluded from essential services."*

Brief level descriptors: - **Level 1 (1)**: one-sided / fragmented (e.g. only "saves money" or only "excludes the elderly"); little development; no conclusion. - **Level 2 (2–3)**: some discussion of both sides but mostly descriptive or one-sided; limited stakeholders; weak or absent conclusion. - **Level 3 (4–5)**: balanced, several stakeholders, **both** benefits and drawbacks with accurate terminology (e.g. *digital divide*) and sustained reasoning, leading to a **justified conclusion** that takes a reasoned position.

Part B (6 marks)

6. [2] 1 mark: Computer Misuse Act 1990. 1 mark: **Offence 2** — unauthorised access *with intent* to commit/facilitate a further offence (here, theft of trade secrets). (*Accept that Offence 1 also occurs, but Offence 2 is the more serious one and is required.*)

7. [2] 1 mark: **TCP** (Transport) splits data into packets, handles reliable delivery — sequencing, acknowledgement and reassembly of packets in the right order. 1 mark: **IP** (Network/Internet) handles addressing and **routing** of packets across networks from source to destination.

8. [2] 1 mark + 1 mark: a multi-core processor has more than one core, so it can execute **multiple threads/instructions truly simultaneously** [1]; tasks that can be split/parallelised are shared across cores and finish sooner (true parallelism rather than time-slicing on one core) [1].

Total: 14 + 6 = 20 marks